

Experiment - 10

16 Bit subtraction

AIM:

To write an assembly language program to implement 16 bit subtraction using 8086 processor.

ALGORITHM:

- 1) Start the program by loading a register pair with address of first number.
- 2) copy the data to another register pair.
- 3) load the second number to first register pair.
- 4) subtract the two register pair contents.
- 5) check for borrow.
- 6) store the value of difference + borrow.
- 7) End.

PROGRAM:

Mnemonics	Explanation
MOV AX, [110H]	MOVE A to 1100
MOV BX, [1102H]	MOVE B to 1102
SUB AX, BX	subtract A and B

MOV [1200H],
AX

HLT

MOV [1200] to
accumulator

Halt.

INPUT:

ADDRESS	DATA
1100	30
1102	15

OUTPUT:

ADDRESS	DATA
1200	15

RESULT:

o/p 2

Thus the program was
executed successfully using 8086
simulator.

Saved to this PC

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step delay ms: 0
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0100:000C